

Problem

If an ammeter with an internal resistance of $100\ \Omega$ and a current capacity of $2\ \text{mA}$ is to measure $5\ \text{A}$, determine the value of the resistance needed. Calculate the power dissipated in the shunt resistor.

Step-by-step solution

Step 1 of 3

The internal resistance of ammeter, $R_m = 100\ \Omega$

Its current capacity, $I_m = 2\ \text{mA}$

The current to be measured, $I = 5\ \text{A}$

Draw the ammeter circuit with shunt resistance.

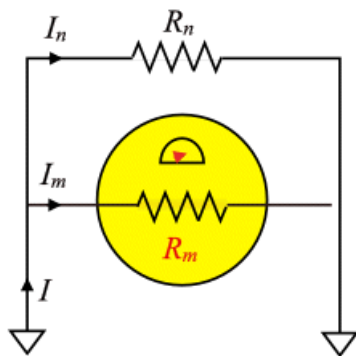


Figure 1

Step 2 of 3

From the circuit in Figure 1, determine the current through the shunt resistance, I_n .

$$\begin{aligned} I_n &= I - I_m \\ &= 5 - 2 \times 10^{-3} \\ &= 4.998\ \text{A} \end{aligned}$$

Determine the shunt resistance, R_n .

The voltage across resistance R_n is equal to resistance R_m .

$$\begin{aligned} R_n I_n &= I_m R_m \\ R_n &= \frac{I_m}{I_n} R_m \\ &= \left(\frac{2 \times 10^{-3}}{4.998} \right) (100) \\ &= 0.04\ \Omega \end{aligned}$$

Thus, the shunt resistance needed is $\boxed{0.04\ \Omega}$.

Step 3 of 3

Determine the power dissipated in the shunt resistor.

$$\begin{aligned}P_n &= I_n^2 R_n \\&= (4.998^2)(0.04) \\&= 0.999 \\&\approx 1 \text{ W}\end{aligned}$$

Thus, the power dissipated in the shunt resistor is **1 W**.